Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 1 / 方法 1 [AL08-M01]

Trigonometric algebra becomes reliable when each substitution carries its angle range. Keep complementary and supplementary branches, denominator signs, endpoint exclusions, and the attainability of any equality case.

三角代数是否可靠，取决于每次代换是否保留角的范围。补角分支、分母符号、端点排除及等号可达到性都必须保留。

Use a bounded sine-cosine substitution

使用带范围的正余弦代换

Procedure

1. Translate the original domain into allowed signs of sine and cosine.

2. Use $\sin^2 x + \cos^2 x = 1$ to reduce the equation to one bounded variable.

3. Solve the polynomial or inequality, then restore the unsquared signs and any logarithm domain.

4. Use double-angle identities to recover the target rather than solving for the angle itself.

通用步骤

1. 先将原定义域转为正余弦的允许符号。

2. 用平方和恒等式将方程化为一个有界变量。

3. 解多项式或不等式后，恢复未平方符号及对数定义域。

4. 用倍角公式恢复目标，不必求角本身。

> Example. If $0 < x < \pi/4$ and $\sin x \cos x = 1/5$, find $\sin 4x$.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 1 / 方法 1 [AL08-M01] (continued / 续)

> 例题：若 $0 < x < \pi/4$ 且 $\sin x \cos x = 1/5$，求 $\sin 4x$。

Worked solution

1. The double-angle formula gives $\sin 2x = 2/5$.

2. Since $0 < 2x < \pi/2$, the positive cosine is $\cos 2x = \sqrt{1 - 4/25} = \sqrt{21}/5$.

3. Thus $\sin 4x = 2 \sin 2x \cos 2x = 4\sqrt{21}/25$. The angle interval determined the radical sign.

例题解析

1. 倍角公式给出 $\sin 2x = 2/5$。

2. 由 $0 < 2x < \pi/2$，余弦取正，故 $\cos 2x = \sqrt{21}/5$。

3. 因此 $\sin 4x = 4\sqrt{21}/25$，根式符号由角区间确定。

Squaring removes signs, so every algebraic root must return to the original domain.

平方会丢失符号，所有代数根都须返回原定义域检查。

Practice / 配套练习: AIME-2003-I-Q04 (WB p. 100), AIME-2003-I-Q11 (WB p. 773), AIME-2006-I-Q08 (WB p. 250), AIME-2011-I-Q09 (WB p. 119), AIME-2012-II-Q09 (WB p. 195), AIME-2023-I-Q08 (WB p. 381)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 2 / 方法 2 [AL08-M02]

List all trigonometric branches over a finite interval

在有限区间列全三角分支

Procedure

1. Use both exact families: $\sin u = \sin v$ gives $u - v = 2k\pi$ or $u + v = (2k+1)\pi$; $\cos u = \cos v$ gives $u = \pm v + 2k\pi$.

2. Intersect each arithmetic angle progression with the stated open or closed interval.

3. Count duplicates by solving the simultaneous progression equations.

4. For a sine level, treat zero, interior levels, and turning-point extrema separately.

通用步骤

1. 列全准确分支：$\sin u = \sin v$ 时，$u - v = 2k\pi$ 或 $u + v = (2k+1)\pi$；$\cos u = \cos v$ 时，$u = \pm v + 2k\pi$。

2. 将每个等差角数列与给定开闭区间相交。

3. 联立两数列方程，检查重复解。

4. 数正弦水平交点时，零、内部水平及转折极值须分别处理。

> Example. How many $x \in [0, 2\pi)$ satisfy $\cos 3x = \sin 2x$?

> 例题：在 $[0, 2\pi)$ 内，有多少个 x 满足 $\cos 3x = \sin 2x$?

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 2 / 方法 2 [AL08-M02] (continued / 续)

Worked solution

1. Write $\sin 2x = \cos(\pi/2 - 2x)$. Equal cosines give $3x = \pi/2 - 2x + 2k\pi$ or $3x = -\pi/2 + 2x + 2k\pi$.

2. The first family is $x = \pi/10 + 2k\pi/5$, with five representatives; the second is $x = -\pi/2 + 2k\pi$, with only $x = 3\pi/2$.

3. That second value is not in the first family, since it would require $k = 7/2$. Hence there are six distinct solutions.

例题解析

1. 将右边写为 $\cos(\pi/2 - 2x)$，余弦相等给出两个差角分支。

2. 第一支为 $x = \pi/10 + 2k\pi/5$，区间内五个；第二支为 $x = -\pi/2 + 2k\pi$，仅有 $3\pi/2$。

3. 后一值若属于第一支，会要求 $k = 7/2$，非整数，因此共有六个不同解。

Equal sine does not mean equal angle; the supplementary family is often decisive.

正弦相等不代表角相等，补角支往往决定答案。

The full rules are $\sin u = \sin v$ iff $u - v = 2k\pi$ or $u + v = (2k+1)\pi$, and $\cos u = \cos v$ iff $u = \pm v + 2k\pi$. For a nested sine, split its inner argument into monotone arcs before counting levels; a zero derivative at an inner extremum can be a tangency rather than two crossings.

For a periodic sign itinerary under angle multiplication, shifting one itinerary period preserves the full condition. If uniqueness is stated, the reduced angle is fixed by that shift; solve its finite congruence candidates and verify every sign in one complete period.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 2 / 方法 2 [AL08-M02] (continued / 续)

完整规则为：正弦相等当且仅当 $u - v = 2k\pi$ 或 $u + v = (2k+1)\pi$；余弦相等当且仅当 $u = \pm v + 2k\pi$。嵌套正弦数水平交点时，先把内部角分成单调弧；内部极值的零导数可能对应相切，而不是两次穿越。

角倍增下的周期符号轨迹，移动一个信息周期后仍满足完整条件；若已知唯一性，化简后的角须为该移动的不动点。解有限同余候选，再核对一整个符号周期。

Practice / 配套练习: AIME-2002-II-Q10 (WB p. 192), AIME-2018-I-Q06 (WB p. 81), AIME-2019-II-Q10 (WB p. 208), AIME-2024-I-Q12 (WB p. 179), AIME-2025-II-Q09 (WB p. 197)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 3 / 方法 3 [AL08-M03]

Factor with sum-to-product before dividing

先和差化积再考虑相除

Procedure

1. Pair symmetric angle multiples with sum-to-product identities.

2. Look for a common factor or a cubic identity rather than expanding every term.

3. Keep the zero-factor branches; divide only after handling them separately.

4. Use the triangle angle sum to simplify cyclic expressions when applicable.

通用步骤

1. 将对称倍角用和差化积配对。

2. 寻找公共因子或立方恒等式，不必逐项展开。

3. 保留因子为零的分支，单独处理后才能相除。

4. 涉及三角形循环式时，利用角和进一步化简。

> Example. Solve $\cos x + \cos 3x = \cos 2x$ for $0 < x < \pi$.

> 例题：在 $0 < x < \pi$ 内解 $\cos x + \cos 3x = \cos 2x$。

Worked solution

1. The left side is $2\cos 2x \cos x$, so the equation factors as $\cos 2x(2\cos x - 1) = 0$.

2. The first factor gives $x = \pi/4, 3\pi/4$; the second gives $x = \pi/3$.

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三角恒等式与方程

Method 3 / 方法 3 [AL08-M03] (continued / 续)

3. All three are in the interval and distinct. Dividing at the start by $\cos 2x$ would have lost two roots.

例题解析

1. 左边为 $2\cos 2x\cos x$，所以方程因式分解为 $\cos 2x(2\cos x-1)=0$。

2. 第一因子给出 $\pi/4, 3\pi/4$，第二因子给出 $\pi/3$。

3. 三根均在区间内且互异，若一开始除以 $\cos 2x$，就会漏去两根。

A factorized identity exposes exceptional branches that an early division hides.

因式分解能揭示过早相除所遮住的特殊分支。

For triangle angles, the sine and cosine laws give $\sin^2 A+\sin^2 B-2\sin A\sin B\cos C=\sin^2 C$. Therefore $\cos^2 A+\cos^2 B+2\sin A\sin B\cos C=2-\sin^2 C$. Also $1-\sum\cos 3A=4\sin(3A/2)\sin(3B/2)\sin(3C/2)$.

Products under a fixed angle sum use $2\cos u\cos v=\cos(u-v)+\cos(u+v)$ and $\sin u+\sin v=2\sin((u+v)/2)\cos((u-v)/2)$. A formal maximum of the resulting cosine must still satisfy all original side-segment constraints.

三角形的正余弦定理给出 $\sin^2 A+\sin^2 B-2\sin A\sin B\cos C=\sin^2 C$，故对应余弦平方循环式等于 $2-\sin^2 C$；另有 $1-\sum\cos 3A=4\sin(3A/2)\sin(3B/2)\sin(3C/2)$。

固定角和下的积可用 $2\cos u\cos v=\cos(u-v)+\cos(u+v)$，和可用正弦和差化积。所得余弦的形式最大值，仍须符合原来的边线段容纳条件。

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 3 / 方法 3 [AL08-M03] (continued / 续)

Practice / 配套练习: AIME-2006-I-Q12 (WB p. 199), AIME-2013-II-Q15 (WB p. 200), AIME-2014-II-Q12 (WB p. 201), AIME-2016-I-Q09 (WB p. 183), AIME-2025-II-Q14 (WB p. 256)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 4 / 方法 4 [AL08-M04]

Add tangents with full angle-branch control

带完整角分支地相加正切

Procedure

1. Track the ranges of the angles before applying tangent addition.

2. Use $\tan(u+v)=(\tan u+\tan v)/(1-\tan u\tan v)$ with separate handling of a zero denominator.

3. For tangent polygons, express each vertex half-angle as $\arctan(r/t)$ and impose their actual total.

4. Filter algebraic roots using monotonicity of the full angle sum, not merely its tangent.

通用步骤

1. 使用正切和角前先跟踪各角范围。

2. 应用和角公式，分母为零时须另行处理。

3. 切圆多边形中，各顶点半角可写为 $\arctan(r/t)$，并保留真实总角。

4. 利用整个角和的单调性筛选代数根，而非只看正切是否相等。

> Example. Find the exact angle $\arctan 2+\arctan 3$.

> 例题：求 $\arctan 2+\arctan 3$ 的精确值。

Worked solution

1. Both angles lie in $(0,\pi/2)$, so the sum lies in $(0,\pi)$.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 4 / 方法 4 [AL08-M04] (continued / 续)

2. The tangent is $(2+3)/(1-6) = -1$. The negative tangent places the sum in the second quadrant.

3. Therefore the sum is $3\pi/4$, not the principal arctangent $-\pi/4$.

例题解析

1. 两角均在第一象限，和在零与 π 之间。

2. 正切为 $(2+3)/(1-6) = -1$，负号说明角和在第二象限。

3. 因此和为 $3\pi/4$，而不是主值反正切 $-\pi/4$。

Tangent identifies an angle only modulo π; the original angle total chooses the branch.

正切只能确定模 π 的角，须由原角和选择分支。

A convex n-gon has half-angle total $(n-2)\pi/2$. Each positive term $\arctan(r/t_i)$ increases strictly with r, so the radius satisfying this full total is unique. A root of a tangent equation may instead represent another total differing by π.

For a radical sine recurrence, $\sqrt{1-\sin^2\phi} = |\cos\phi|$. Crossing $\pi/2$ changes the sign of the cosine term and can fold an increasing angle sequence into a short cycle.

凸 n 边形半角和为 $(n-2)\pi/2$。每个正项 $\arctan(r/t_i)$ 随 r 严格增加，故满足完整角和的半径唯一；正切方程的某个根可能对应相差 π 的另一个总角。

根式正弦递推中，$\sqrt{1-\sin^2\phi} = |\cos\phi|$；跨越直角会改变余弦项符号，将递增形式角折叠成短循环。

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三角恒等式与方程

Method 4 / 方法 4 [AL08-M04] (continued / 续)

Practice / 配套练习: AIME-2000-II-Q10 (WB p. 331), AIME-2002-II-Q15 (WB p. 341), AIME-2008-I-Q08 (WB p. 193), AIME-2009-II-Q14 (WB p. 213), AIME-2016-II-Q14 (WB p. 433), AIME-2020-II-Q13 (WB p. 347)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 5 / 方法 5 [AL08-M05]

Use a two-root recurrence for high trigonometric powers

以二根递推处理高次三角幂

Procedure

1. Put $u=\sin^2 x, v=\cos^2 x$, so $u+v=1$ and $p=uv\in[0,1/4]$.

2. The power sums satisfy $S_0=2, S_1=u+v$ and $S_n=(u+v)S_{n-1}-uvS_{n-2}$.

3. Use the allowed product interval to reject extraneous algebraic roots.

4. When irrational coefficients occur, prove the rational/irrational parity pattern by induction before studying last digits.

通用步骤

1. 令 $u=\sin^2 x, v=\cos^2 x$，则和为一，积在零至四分之一之间。

2. 幂和满足 $S_n=(u+v)S_{n-1}-uvS_{n-2}$，初值为二及 $u+v$。

3. 使用乘积允许区间排除多余代数根。

4. 若递推系数含无理数，先归纳证明有理、无理的下标奇偶规律，再研究末位。

> Example. If $\cos 2x=1/3$, find $\sin^8 x+\cos^8 x$ using a power-sum recurrence.

> 例题：若 $\cos 2x=1/3$，用幂和递推求 $\sin^8 x+\cos^8 x$。

Worked solution

1. Here $u+v=1$ and $uv=(1-\cos^2 2x)/4=2/9$.

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三角恒等式与方程

Method 5 / 方法 5 [AL08-M05] (continued / 续)

2. The recurrence gives $S_2 = 1 - 4/9 = 5/9$ and $S_3 = 5/9 - 2/9 = 1/3$.

3. Thus $S_4 = 1/3 - (2/9)(5/9) = 17/81$, the required eighth-power sum.

例题解析

1. 此时 $u+v=1, uv=(1-\cos^2 2x)/4=2/9$。

2. 递推得 $S_2=5/9, S_3=1/3$。

3. 所以 $S_4=1/3-(2/9)(5/9)=17/81$，即所求八次幂和。

An integer subsequence needs an integrality proof before modular arithmetic applies.

只有先证明子序列为整数，才能对它作整数同余。

Practice / 配套练习: AIME-2019-I-Q08 (WB p. 51), AIME-2023-II-Q13 (WB p. 205)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 6 / 方法 6 [AL08-M06]

Telescope trigonometric sums by endpoint differences

将三角和望远镜化为端点差

Procedure

1. Use product-to-sum or a cotangent difference to express each summand as a discrete difference.

2. Write the finite sum with explicit first and last terms before canceling.

3. For alternating samples, pair supplementary angles and account for unpaired endpoints.

4. Translate the final sign or integer-value condition into a periodic angle interval.

通用步骤

1. 用积化和差或余切差，将每项变成离散差分。

2. 消去之前，明确写出有限和首尾项。

3. 交替采样时配对互补角，并保留未配对端点。

4. 把最终符号或整数值要求转为周期角区间。

> Example. Find a closed form for $\sum_{k=1}^{n} \sin((2k-1)t)$ when $\sin t \neq 0$.

> 例题：当 $\sin t \neq 0$，求 $\sum_{k=1}^{n} \sin((2k-1)t)$ 的闭式。

Worked solution

1. The identity $2\sin t \sin((2k-1)t) = \cos((2k-2)t) - \cos(2kt)$ gives a difference.

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三角恒等式与方程

Method 6 / 方法 6 [AL08-M06] (continued / 续)

2. Summing cancels every interior cosine, leaving $1-\cos 2nt$.

3. Divide by $2\sin t$ and use $1-\cos 2nt = 2\sin^2 nt$. The answer is $\sin^2(nt)/\sin t$.

例题解析

1. 由积化和差，每项乘 $2\sin t$ 后为两个相邻偶倍角余弦之差。

2. 求和时内部余弦全部抵消，只余 $1-\cos 2nt$。

3. 除以 $2\sin t$，再用倍角公式，得闭式 $\sin^2(nt)/\sin t$。

The identity $\cot u - \cot v = \sin(v-u)/(\sin u \sin v)$ is a useful reciprocal-product version.

余切差恒等式是处理正弦乘积倒数的有用版本。

Practice / 配套练习: AIME-2000-II-Q15 (WB p. 198), AIME-2008-II-Q08 (WB p. 194), AIME-2015-II-Q13 (WB p. 203)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 7 / 方法 7 [AL08-M07]

Pair and double sine products, or use roots of unity

配对倍角正弦积或使用单位根

Procedure

1. Pair complementary or supplementary angles to relate two products.

2. Apply $2\sin x\cos x=\sin 2x$, then match the doubled-angle multiset exactly.

3. Cancel only nonzero products and preserve any central factor equal to one.

4. Alternatively factor a root-of-unity polynomial at one and take magnitudes.

通用步骤

1. 将余角或补角配对，建立两个乘积的关系。

2. 使用正弦倍角公式，准确匹配倍角后的角多重集。

3. 只约去非零乘积，并保留中央等于一的因子。

4. 也可在一处计算单位根多项式的因式分解，再取模。

> Example. Prove $\sin(\pi/5)\sin(2\pi/5)=\sqrt{5}/4$.

> 例题：证明 $\sin(\pi/5)\sin(2\pi/5)=\sqrt{5}/4$。

Worked solution

1. Let $\zeta=e^{2\pi i/5}$. Evaluating $(x^5-1)/(x-1)$ at one gives $\prod_{k=1}^{4}(1-\zeta^k)=5$.

2. Taking magnitudes and using $|1-e^{it}|=2|\sin(t/2)|$ gives $16\prod_{k=1}^{4}\sin(k\pi/5)=5$.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 7 / 方法 7 [AL08-M07] (continued / 续)

3. Supplementary sines pair equally, so the product is $(\sin(\pi/5)\sin(2\pi/5))^2 = 5/16$. Both factors are positive, giving the stated positive square root.

例题解析

1. 令五次单位根为 ζ，在一处计算 $(x^5-1)/(x-1)$，得四个因子的乘积为五。

2. 取模并用弦长公式，得到 $16\prod_{k=1}^{4}\sin(k\pi/5) = 5$。

3. 补角正弦相等，因此所求积的平方为 $5/16$；两因子均正，故取正根 $\sqrt{5}/4$。

A product identity depends on the entire angle multiset, not only its number of factors.

乘积恒等式依赖整个角多重集，而不只依赖因子数量。

Practice / 配套练习: AIME-2015-I-Q13 (WB p. 202)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 8 / 方法 8 [AL08-M08]

Convert multiple-angle metric relations into polynomial identities

把倍角度量关系转为多项式恒等式

Procedure

1. Use the geometry to choose one angle and prove its range before algebra.

2. Apply the chord formula $2R\sin(\phi/2)$ or the sine rule to express length ratios.

3. Use double- and triple-angle identities to eliminate the angle in favor of a bounded sine or cosine.

4. Check every root against the geometric ordering and any supplementary-angle choices.

通用步骤

1. 先由几何选择一个角，并证明它的范围。

2. 使用弦长公式或正弦定理表示长度比。

3. 以二倍、三倍角恒等式，将角消为有界正弦或余弦。

4. 按几何顺序及可能的补角选择筛选每个根。

> Example. Prove $\sin 3t/\sin t = 1 + 2\cos 2t$ for $0 < t < \pi$.

> 例题：对 $0 < t < \pi$，证明 $\sin 3t/\sin t = 1 + 2\cos 2t$。

Worked solution

1. Addition gives $\sin 3t = \sin t\cos 2t + \cos t\sin 2t$.

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三角恒等式与方程

Method 8 / 方法 8 [AL08-M08] (continued / 续)

2. Use $\sin 2t = 2\sin t\cos t$ and divide by the positive $\sin t$, obtaining $\cos 2t + 2\cos^2 t$.

3. Since $2\cos^2 t = 1 + \cos 2t$, this equals $1 + 2\cos 2t$, also $4\cos^2 t - 1$.

例题解析

1. 和角公式给出 $\sin 3t = \sin t\cos 2t + \cos t\sin 2t$。

2. 代入二倍角并除以正的 $\sin t$，得 $\cos 2t + 2\cos^2 t$。

3. 再由余弦倍角恒等式，化为 $1 + 2\cos 2t = 4\cos^2 t - 1$。

A chord may correspond to a major consecutive arc; exclude or include that case from the polygon geometry.

一条边可能对应连续的优弧，须由多边形构型决定是否排除。

Practice / 配套练习: AIME-2001-I-Q13 (WB p. 292), AIME-2010-II-Q14 (WB p. 279), AIME-2011-I-Q14 (WB p. 394), AIME-2013-II-Q08 (WB p. 285), AIME-2018-I-Q15 (WB p. 300)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 9 / 方法 9 [AL08-M09]

Encode geometric angles by dot products and equally turning vectors

用点积及等转角向量表示几何角

Procedure

1. Write each direction as a unit sine-cosine vector, keeping orientation.

2. Use dot products for cosine and determinants for signed sine; a tangent ratio also needs the dot-product sign.

3. For a closed equiangular polygon, the weighted direction vectors sum to zero.

4. Derive exact special-angle cosines from the closure or root-of-unity relation when needed.

通用步骤

1. 将方向写成带定向的单位正余弦向量。

2. 点积给余弦，行列式给有向正弦；求正切还须保留点积符号。

3. 闭合等角多边形的加权边方向向量和为零。

4. 需要时由闭合或单位根关系推导特殊角余弦的精确值。

> Example. Derive exact values for $\cos 72^\circ$ and $\cos 144^\circ$ from a regular pentagon.

> 例题：由正五边形推导 $\cos 72^\circ$ 和 $\cos 144^\circ$ 的精确值。

Worked solution

1. The five equally spaced unit vectors sum to zero. Real parts give $1 + 2\cos 72^\circ + 2\cos 144^\circ = 0$.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 9 / 方法 9 [AL08-M09] (continued / 续)

2. Put $c = \cos 72^\circ$ and use $\cos 144^\circ = 2c^2 - 1$. This gives $4c^2 + 2c - 1 = 0$.

3. Since 72° is acute, $c = (\sqrt{5} - 1)/4$. Substitution gives $\cos 144^\circ = -(\sqrt{5} + 1)/4$.

例题解析

1. 五个等距单位方向向量和为零，取实部得 $1 + 2\cos 72^\circ + 2\cos 144^\circ = 0$。

2. 令 $c = \cos 72^\circ$，用倍角公式得 $4c^2 + 2c - 1 = 0$。

3. 由七十二度为锐角，取正根 $c = (\sqrt{5} - 1)/4$，继而得 $\cos 144^\circ = -(\sqrt{5} + 1)/4$。

For rolling, ground-frame spin includes rotation of the line of centers as well as relative contact rotation.

滚动的对地自转还包括圆心连线的转动，不能只算相对接触转角。

Practice / 配套练习: AIME-2005-II-Q12 (WB p. 392), AIME-2014-I-Q10 (WB p. 374), AIME-2026-I-Q14 (WB p. 414)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 10 / 方法 10 [AL08-M10]

Linearize complementary radicals with sine squares

以正弦平方线性化互补根式

Procedure

1. Use the real-radical domain to bound each variable, then put it equal to a scaled sine square.

2. Choose angles in $[0, \pi/2]$ so all square roots have the intended positive signs.

3. The paired radicals become a sine of an angle sum.

4. List both supplementary pair-sum branches, solve their linear angle equations, and reject those outside the allowed range.

通用步骤

1. 由实根式定义域给变量界，再令其等于适当倍数的正弦平方。

2. 选角在第一象限闭区间内，保证所有根式取正确非负符号。

3. 成对互补根式会变成角和的正弦。

4. 列出角和的两种补角支，解线性角方程，并排除超范围分支。

> Example. For $0 \le x, y \le 1$, characterize solutions of $\sqrt{x(1-y)} + \sqrt{y(1-x)} = 1$.

> 例题：对 $0 \le x, y \le 1$，刻画 $\sqrt{x(1-y)} + \sqrt{y(1-x)} = 1$ 的全部解。

Worked solution

1. Put $x = \sin^2 a, y = \sin^2 b$ with $a, b \in [0, \pi/2]$. The left side is $\sin(a+b)$.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 10 / 方法 10 [AL08-M10] (continued / 续)

2. As $a+b \in [0,\pi]$, equality to one requires $a+b=\pi/2$.

3. Thus $y=\cos^2 a = 1-x$. Conversely every such pair satisfies the original equation, so the complete solution is $x+y=1$ in the given square.

例题解析

1. 令 $x=\sin^2 a, y=\sin^2 b$，角均在零至直角之间，左边为 $\sin(a+b)$。

2. 角和在零至 π 之间，正弦为一只能是角和为直角。

3. 所以 $y=1-x$；反之该关系确满足原方程，故全部解为给定方形内的 $x+y=1$。

> The angle range is what makes the radical identity exact without extra absolute values.

> 角范围保证根式恒等式准确成立，不额外产生绝对值。

Practice / 配套练习: AIME-2022-I-Q15 (WB p. 204)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 11 / 方法 11 [AL08-M11]

Join sine and cosine series into a complex geometric series

将正余弦级数合成复等比级数

Procedure

1. Express phase shifts through powers of e^{it} or ie^{it}.

2. Combine the two real series as real and imaginary parts of one absolutely convergent geometric series.

3. Sum it exactly, rationalize once, then read both parts.

4. Restore quadrant and denominator conditions after any squared equation.

通用步骤

1. 以 e^{it} 或 ie^{it} 的幂表示相位变化。

2. 将两条实级数合为一个绝对收敛复等比级数的实虚部。

3. 精确求和并统一有理化，再读出两部分。

4. 若后续平方求解，须恢复象限及分母条件。

> Example. Find $\sum_{k=1}^{\infty} 3^{-k}\sin(k\pi/3)$.

> 例题：求 $\sum_{k=1}^{\infty} 3^{-k}\sin(k\pi/3)$。

Worked solution

1. The series is the imaginary part of $\sum_{k\geq 1}(e^{i\pi/3}/3)^k$, whose ratio modulus is $1/3$.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 11 / 方法 11 [AL08-M11] (continued / 续)

2. For general $|r|<1$, that imaginary part is $r\sin t/(1-2r\cos t+r^2)$, by rationalizing $re^{it}/(1-re^{it})$.

3. With $r=1/3, t=\pi/3$, it is $(\sqrt{3}/6)/(7/9)=3\sqrt{3}/14$.

例题解析

1. 所求为公比 $e^{i\pi/3}/3$ 的复等比级数虚部，公比模小于一。

2. 在公比满足 $|r|<1$ 时，将闭式有理化，一般虚部为 $r\sin t/(1-2r\cos t+r^2)$。

3. 代入 $r=1/3, t=\pi/3$，得 $(\sqrt{3}/6)/(7/9)=3\sqrt{3}/14$。

Absolute convergence permits joining the series; it should be stated before rearranging.

应先说明绝对收敛，再合并或重排级数。

Practice / 配套练习: AIME-2013-I-Q14 (WB p. 93)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 12 / 方法 12 [AL08-M12]

Check compatibility of simultaneous trigonometric maxima

检查多个三角项能否同时取最大

Procedure

1. If a sum attains the sum of individual upper bounds, every summand must attain its own bound.

2. Translate those equalities to angular congruences and test compatibility using gcd or Bezout.

3. If the bounds are incompatible, reduce the actual expression using an identity and optimize on its true range.

4. Exhibit an angle realizing the final compatible equality case.

通用步骤

1. 和要达到各项上界之和，每项必须同时达到自身上界。

2. 将取等条件转为角同余，并用最大公因数或贝祖检验相容性。

3. 若不相容，利用恒等式在真实范围内优化原表达式。

4. 给出实现最后取等条件的角。

> Example. Find the maximum of $\sin 2x + \cos 4x$ for real x.

> 例题：求实数域上 $\sin 2x + \cos 4x$ 的最大值。

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 12 / 方法 12 [AL08-M12] (continued / 续)

Worked solution

1. The separate upper bounds sum to two, but $\sin 2x = 1$ forces $\cos 4x = -1$, so they cannot both attain one.

2. Put $u = \sin 2x \in [-1, 1]$. Then the expression is $1 + u - 2u^2 = 9/8 - 2(u - 1/4)^2$.

3. The value $u = 1/4$ is attainable by a real angle, so the exact maximum is $9/8$.

例题解析

1. 各项上界相加为二，但 $\sin 2x = 1$ 会迫使 $\cos 4x = -1$，不能同时为一。

2. 令 $u = \sin 2x \in [-1, 1]$，表达式为 $1 + u - 2u^2 = 9/8 - 2(u - 1/4)^2$。

3. 实角可实现 $u = 1/4$，所以准确最大值为 $9/8$。

An algebraic upper bound is not a maximum until all its equality requirements are compatible.

只有全部取等要求相容，代数上界才能称为最大值。

For positive frequencies $m = gu, n = gv$ with coprime u, v, simultaneous $\sin(mx) = \sin(nx) = 1$ requires $v(4a + 1) = u(4b + 1)$. Bezout shows integer a, b exist exactly when $u \equiv v \pmod 4$, which forces both odd. This is stronger than merely requiring the two frequencies odd.

将正频率写为 $m = gu, n = gv$，其中 u, v 互质。同时正弦为一要求 $v(4a + 1) = u(4b + 1)$。由贝祖，整数 a, b 存在当且仅当 $u \equiv v \pmod 4$，从而两者均奇；仅要求两频率为奇数还不够。

Practice / 配套练习: AIME-2021-I-Q07 (WB p. 196)